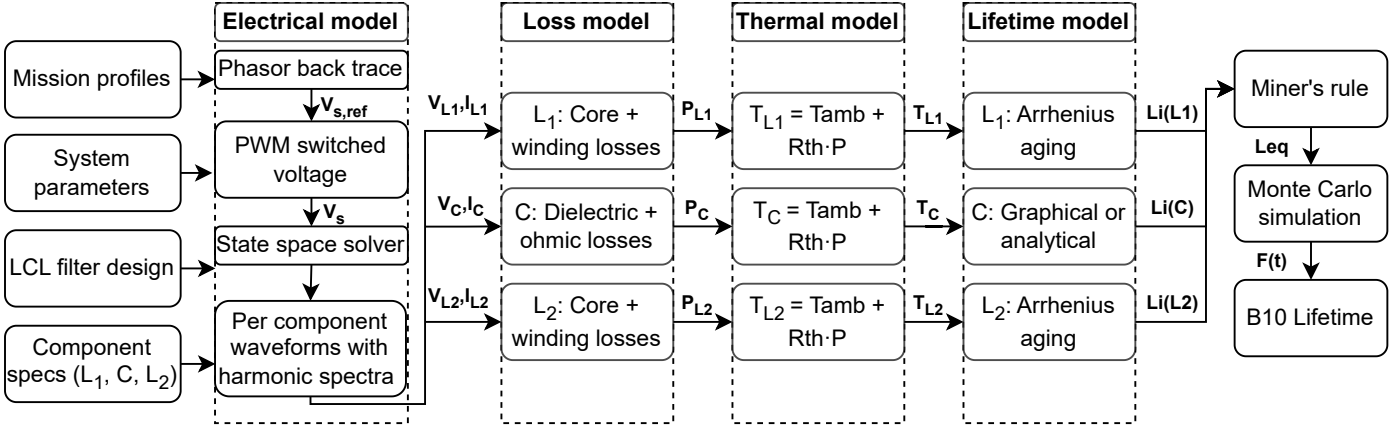

Electrical model
Loss model
Thermal model
Lifetime model
Mission profiles
System parameters
LCL filter design
Component specs (L1, C, L2)
Phasor back trace
Vs,ref
PWM switched voltage
Vs
State space solver
Per component waveforms with harmonic spectra
VL1,IL1
VC,IC
VL2,IL2
L1: Core + winding losses
C: Dielectric + ohmic losses
L2: Core + winding losses
PL1
PC
PL2
TL1 = Tamb + Rth·P
TC = Tamb + Rth·P
TL2 = Tamb + Rth·P
TL1
TC
TL2
L1: Arrhenius aging
C: Graphical or analytical
L2: Arrhenius aging
Li(L1)
Li(C)
Li(L2)
Miner's rule
Leq
Monte Carlo simulation
F(t)
B10 Lifetime